

Information Technologies for Industrial Engineers

เทคโนโลยีสารสนเทศสำหรับวิศวกรอุตสาหกรรม

Decentralized Application (DApp)

Demo

<https://nr-dapp-68.netlify.app/>

What you need

- Alchemy API key
- Alchemy URL
- Addresses
 - Account address (from MetaMask)
 - Secret contract address
 - ERC20 (token) contract address
- Token Symbol

What you need

- ABIs
 - Secret contract address
 - ERC20 (token) contract address

Contract codes

- `secret.sol`
- `token.sol`

Setup

- `pnpm create vite@latest`
- ...
- ...
- `pnpm install wagmi viem@2.x @tanstack/react-query`
- `pnpm approve-builds`

.env

```
VITE_ALCHEMY_KEY=  
VITE_ALCHEMY_URL=  
VITE_MY_ACCOUNT_ADDRESS=  
VITE_SECRET_CONTRACT_ADDRESS=  
VITE_TOKEN_CONTRACT_ADDRESS=  
VITE_TOKEN_SYMBOL=
```


Example `.env`

```
VITE_ALCHEMY_URL=  
VITE_ALCHEMY_KEY=  
VITE_MY_ACCOUNT_ADDRESS=0x6269f27234747F4ac12A6d5E88c75e021da290CF  
VITE_SECRET_CONTRACT_ADDRESS=0xFC76182992cB0690Ca61742eB07246EB1a7705FE  
VITE_TOKEN_CONTRACT_ADDRESS=0x654757D6a4F2291DF64Da93dc078dF6fF7A2d29D  
VITE_TOKEN_SYMBOL=POH52
```

`./src/env.ts`

See code on the next page

```
export const ALCHEMY_URL = import.meta.env.VITE_ALCHEMY_URL;
export const ALCHEMY_KEY = import.meta.env.VITE_ALCHEMY_KEY;
export const MY_ACCOUNT_ADDRESS = import.meta.env.VITE_MY_ACCOUNT_ADDRESS;
export const SECRET_CONTRACT_ADDRESS = import.meta.env
  .VITE_SECRET_CONTRACT_ADDRESS;
export const TOKEN_CONTRACT_ADDRESS = import.meta.env
  .VITE_TOKEN_CONTRACT_ADDRESS;
export const TOKEN_SYMBOL = import.meta.env.VITE_TOKEN_SYMBOL;

if (
  !ALCHEMY_URL ||
  !ALCHEMY_KEY ||
  !MY_ACCOUNT_ADDRESS ||
  !SECRET_CONTRACT_ADDRESS ||
  !TOKEN_CONTRACT_ADDRESS ||
  !TOKEN_SYMBOL
) {
  throw new Error("Please set env variables correctly.");
}
```

./index.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/vite.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <link
      rel="stylesheet"
      href="https://cdn.jsdelivr.net/npm/@picocss/pico@2/css/pico.lime.min.css"
    />
    <title>DApp</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.tsx"></script>
  </body>
</html>
```

`./src/main.tsx`

| See code on the next page

```

import { QueryClient, QueryClientProvider } from "@tanstack/react-query";
import React from "react";
import ReactDOM from "react-dom/client";
import { createConfig, http, WagmiProvider } from "wagmi";
import { sepolia } from "wagmi/chains";
import App from "./App.tsx";
import { ALCHEMY_KEY, ALCHEMY_URL } from "./env.ts";

const queryClient = new QueryClient();

const config = createConfig({
  chains: [sepolia],
  transports: {
    [sepolia.id]: http(ALCHEMY_URL, {
      name: "Alchemy HTTP Provider",
      key: ALCHEMY_KEY,
    }),
  },
});

ReactDOM.createRoot(document.getElementById("root")!).render(
  <React.StrictMode>
    <WagmiProvider config={config}>
      <QueryClientProvider client={queryClient}>
        <App />
      </QueryClientProvider>
    </WagmiProvider>
  </React.StrictMode>
);

```

`./src/abi.ts`

```
export const secretABI = [  
  // ...  
] as const;  
  
export const tokenABI = [  
  // ...  
] as const;
```

If you don't want to compile the file yourself you may use this file `abi.ts`

./src/App.tsx

<https://gist.github.com/nnnpoooh/47e64662f6667ca1d0d12f4ad359af43#file-app-tsx>